F Structure Prediction

F.1 CASP14 Dataset

For computational efficiency, we filtered the CASP14 dataset to exclude sequences longer than 700 residues to enable single-GPU evaluation. We further filtered out targets where ESMFold predictions showed RMSD > 10 Å compared to the experimental structure, resulting in 17 of the original 34 targets

Data. We downloaded the CASP14 targets dataset from the CASP14 website here.

After filtering, we benchmark on these targets: [T1024, T1025, T1026, T1029, T1030, T1032, T1046s1, T1046s2, T1050, T1054, T1056, T1067, T1073, T1074, T1079, T1082, T1090].

F.2 Sparsity Sweep

We report the difference in RMSD on the above CASP14 dataset between ESMFold and SAE reconstructions across sparsity levels for Matryoshka SAEs. In Figure 7, we see that with only 2^3 to 2^5 active latents, SAEs can reasonably reconstruct structure prediction performance. These results raise questions on how much information is required per token for structure prediction.

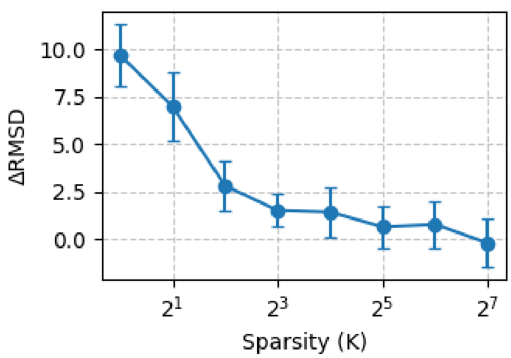

Figure 7: CASP14 ΔRMSD Results Across Sparsity Levels K for Matryoshka SAE Architecture

G Contact Map Prediction

Motivation. Zhang et al. (2024) introduced the Categorical Jacobian calculation to extract coevolutionary signals from protein language models in an unsupervised manner via masked language modeling. Given that SAEs also learn through unsupervised training, we adapt this approach for contact map prediction to evaluate whether SAEs preserve coevolutionary patterns and structural information learned by ESM2.

Data. We randomly sampled 50 proteins from the evaluation dataset in Zhang et al. (2024) due to computational constraints. As shown in Fig. 8, our results replicate the authors' findings, demonstrating superior contact map accuracy across nearly the entire sample. Based on these consistent results, we believe our conclusions would generalize to the complete dataset.

Hyperparameters. In Fig 3b, our SAEs were trained on layer 36 with TopK architecture with expansion factors 16x for ESM2-8M (dict size 10240) and 8x for ESM2-3B (dict size 20480). We did not rerun for the Matryoshka architecture given TopK's similar language modeling reconstruction performance and how computationally expensive it is.

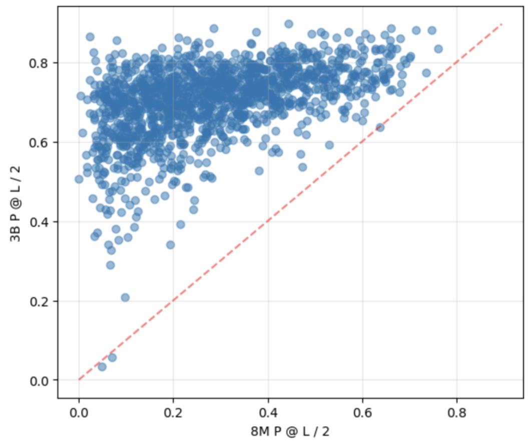

Figure 8: Contact Map Accuracy for Cat. Jacobian on ESM2 8M vs 3B, N=1412/1431

H ESMFold Steering Case Study

H.1 Feature Steering Background

Following the method described in Templeton et al. (2024), each target feature is represented by its corresponding decoder vector ($d_{(i)} = W_{dec}[i, :]$), which is normalized to unit norm during training, and an intervention is performed by adding a scaled version of this vector to the hidden state ($h_l \leftarrow h_l + \alpha \cdot d_{(i)}$) to amplify or suppress that feature's contribution. In our approach, we apply a maximum activation scaling factor during training and subsequently scale the encoder and decoder biases at inference. This means our effective steering coefficient, reported in all figures and results, is given by $\alpha' = \text{norm\_factor} \cdot \alpha$, ensuring consistent steering effects across different model configurations.

H.2 Details of Feature Steering Case Study

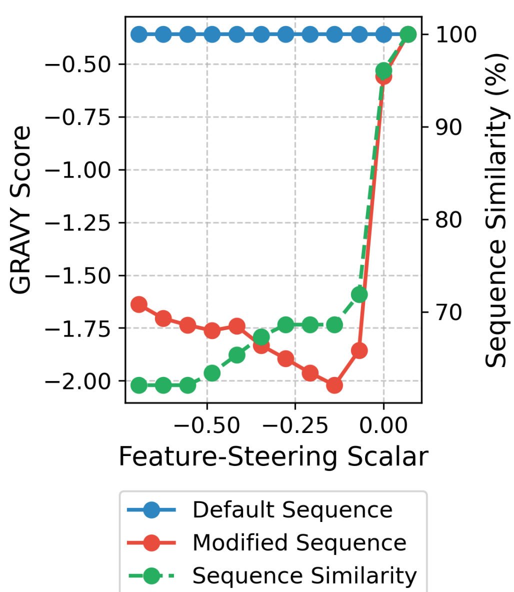

Figure 9: Plot of sequence GRAVY scores overlaid with sequence similarity with increased feature steering

Our intervention framework employed two complementary approaches: (1) examining sequence-level changes through standard ESMFold predictions, and (2) directly intervening on hidden representations while maintaining the true input sequence in an ablated version that isolated intervention effects. This design allowed us to dissect how structural information flows through the model. The ablated model showed minimal performance differences (Figure 2b).

The feature (f/2) was identified in the first group of a Matryoshka SAE trained on hidden layer 36 (dictionary size 10,240, group fractions [0.002, 0.0156, 0.125, 0.8574], sparsity k=20), showing strong correlation with residue hydrophobicity. We selected the Matryoshka SAE architecture as its hierarchical encoding tends to capture broader concepts in earlier groups, which we hypothesized would enable more robust steering across diverse proteins. We applied the steering methodology from Templeton et al. (2024), adding the decoder vector with coefficient α = -0.275. SASA was computed using FreeSASA Mitternacht (2016) with default parameters. The observed RMSD of 2.76 Å aligns with typical deviations for ESMFold with layer 36 LM head ablation, while ablating other ESM2 layer representations resulted in RMSD of 0.191 compared to regular ESMFold (Figure 2b).

As shown in Figure 9, steering progressively decreased sequence GRAVY scores, indicating a shift toward hydrophilic residues. When modified sequences were processed through default ESMFold, we observed even larger SASA increases, consistent with replacing hydrophobic core residues with hydrophilic ones. This effect manifested through two distinct mechanisms: direct biasing of the structure module toward more exposed conformations (even with correct sequence information), and shifting sequence predictions toward more hydrophilic residues when allowed to influence the sequence module. Notably, structural changes persisted even when preserving the original sequence, suggesting ESMFold's predictions are influenced by hidden representations corresponding to the modified sequence despite having access to correct sequence information.

H.3 Validation of Case Study Feature Steering Directionality and Specificity

Scalar	Modified SASA	SASA Change	Modified GRAVY	GRAVY Change
0.069	8369.55	0.00%	-0.3588	0.00%
0.208	8351.58	-0.21%	-0.0595	+83.42%
0.346	8568.43	+2.38%	0.6569	+283.08%
0.554	8616.49	+2.95%	1.3046	+463.60%
0.693	9719.28	+16.13%	1.5255	+525.17%

Table 2: Positive steering control results showing changes in SASA and GRAVY scores relative to baseline values.

Feature	Metric	Scalar Values		
		-0.069	-0.346	-0.693
Feature 5 (Group 1)	δSASA	0.00%	+0.38%	-0.54%
	δGRAVY	0.00%	+40.40%	+52.34%
Feature 39 (Group 2)	δSASA	0.00%	0.00%	0.00%
Feature 1381 (Group 3)	δSASA	0.00%	0.00%	0.00%
Feature 7921 (Group 4)	δSASA	0.00%	0.00%	0.00%
	δGRAVY	0.00%	0.00%	0.00%
Original	δSASA	+128.44%	+137.12%	+112.89%
Experiment	δGRAVY	-417.34%	-446.21%	-356.69%

Table 3: Comparison of random feature controls with original experiment, showing percentage changes relative to unsteered (scalar=0) values.